

Department of Computer Science
Gujarat University
5 YR INT MSC SEM 9 and MSC AIML SEM 3

Subject: Natural Language Processing
Assignment 1 - Practical

- 1) Perform the relevant text preprocessing steps on the dataset provided (use csv, library and text files all) and write your observations/reasons about the preprocessing you did.
- 2) Implement a lemmatizer using the WordNet and compare its output with stemming. Use the Gutenberg Corpus.
- 3) Implement the Bag of Words model to convert a collection of text documents into numerical feature vectors. Use the 20 Newsgroups dataset.
- 4) Implement TFIDF, BOW and One hot Encoding vectorization from scratch using numpy only.
- 5) Implement TF-IDF vectorization and apply it to a text corpus to extract important features. Use the IMDB movie reviews dataset.
- 6) Write a program for sentimental analysis (positive and negative) compare any 3 ML models. Do all the steps of preprocessing which you find relevant to task.
- 7) Write a program to find edit distance between any two string, output should be in pandas data frame
- 8) Write a program for n gram language modelling from scratch using numpy and other relevant modules to predict next word, compare results of unigram, bigram, trigram using perplexity. Store all the n-gram probability in dictionary and then use it for inference.
- 9) Implement Naïve Bayes Classifier for Text classification use dataset of your choice from nltk corpus, preprocess it and use it for classification. Deal all the cases such as out of vocabulary, smoothing etc. Compare the results with library function of naïve bayes using relevant metrics